

Experiment 'p_2_4_b_difference' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_4_b_difference

Natural language statement: null

Method signature: p_2_4_b_difference(x: int, y: int) returns (result: int)

Ensures

- `result == x - y`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_2_4_b_difference(x: int, y: int) returns (result: int)`

The method should respect the following contract:

`ensures result == x - y`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2hcUFS8SjlJ5IKJvTWNDZrGGCHn

Model: gpt-5.5-2026-04-23

Created at: 1780580832

Finish reason: stop

Usage: 252 tokens in, and 43 tokens out

```
{  
  result := x - y;  
}
```

Final Program

```
method p_2_4_b_difference(x: int, y: int) returns (result: int)  
  ensures result == x - y  
{  
  result := x - y;  
}
```

Total Token Usage

Input tokens: 252

Output tokens: 43

Reasoning tokens: 15

Sum of ‘total tokens’: 295

Experiment Timings

Overall Experiment started at 1780580832094, ended at 1780580856200, lasting 24106ms (24.11 seconds)

Iteration #1 started at 1780580832096, ended at 1780580856199, lasting 24103ms (24.10 seconds)